

Problem Set 9 Answers

1. Regardless of the result of Alice's roll, there is a $1/20$ chance that Bob's roll agrees. So the probability is $1/20$ that their rolls agree.

An alternative view: There are 120 equally likely outcomes. Alice's and Bob's results agree on 6 of these outcomes. So the probability is $6/120 = 1/20$.

2. On each round, there are four equally likely possibilities: HH, HT, TH, TT. Consider the very first round on which the result is not TT. On that round, if the result is HH, then they both flipped the coin the same number of times. If the result is HT or TH, then they did not flip the coin the same number of times. So the probability that they both flipped the coin the same number of times is $1/3$.

An alternative view: The probability that they both flipped the coin exactly x times is $(1/2)^{x-1}(1/2)(1/2)^{x-1}(1/2) = (1/2)^{2x} = (1/4)^x$. So the probability that they both flipped the coin the same number of times is $\sum_{x=1}^{\infty} (1/4)^x = \frac{1/4}{1-1/4} = (1/4)/(3/4) = 1/3$.

3a. The probability is $\sum_{y=1}^{\infty} pq^{2y-1} = pq + pq^3 + pq^5 + \dots = pq(1 + q^2 + q^4 + \dots) = pq/(1 - q^2)$.

3b. The probability is $\sum_{y=1}^{\infty} pq^{3y-1} = pq^2 + pq^5 + pq^8 + \dots = pq^2(1 + q^3 + q^6 + \dots) = pq^2/(1 - q^3)$.

3c. The probability is $\sum_{y=1}^{\infty} pq^{4y-1} = pq^3 + pq^7 + pq^{11} + \dots = pq^3(1 + q^4 + q^8 + \dots) = pq^3/(1 - q^4)$.

3d. The probability is $\sum_{y=1}^{\infty} pq^{ry-1} = pq^{r-1} + pq^{2r-1} + pq^{3r-1} + \dots = pq^{r-1}(1 + q^r + q^{2r} + \dots) = pq^{r-1}/(1 - q^r)$.

4. We compute $P(X = Y) = \sum_{x=0}^{\infty} (1/2)^{x+1}(e^{-2})(2^x)/(x!) = (1/2)(e^{-2}) \sum_{x=0}^{\infty} 1/(x!) = (1/2)(e^{-2})(e) = (1/2)(e^{-1})$.

5. First, we note that $P(X \leq Y) = \sum_{x=1}^{\infty} (3/5)^{x-1}(2/5) \sum_{y=x}^{\infty} (2/7)^{y-1}(5/7)$.

We compute $\sum_{y=x}^{\infty} (2/7)^{y-1} = (2/7)^{x-1}/(1 - 2/7) = (2/7)^{x-1}/(5/7)$.

Substituting this back into the original equation yields:

$$P(X \leq Y) = \sum_{x=1}^{\infty} (3/5)^{x-1}(2/5)((2/7)^{x-1}/(5/7))(5/7),$$

which simplifies to:

$$P(X \leq Y) = \sum_{x=1}^{\infty} (6/35)^{x-1}(2/5) = (2/5)/(1 - 6/35) = (2/5)/(29/35) = 14/29.$$